Library Drift in Self-Evolving Skill Libraries

Xing Zhang, Yanwei Cui, Guanghui Wang, Ziyuan Li, Wei Qiu, Bing Zhu, Peiyang He · AWS · HSBC · arXiv 2605.19576v3

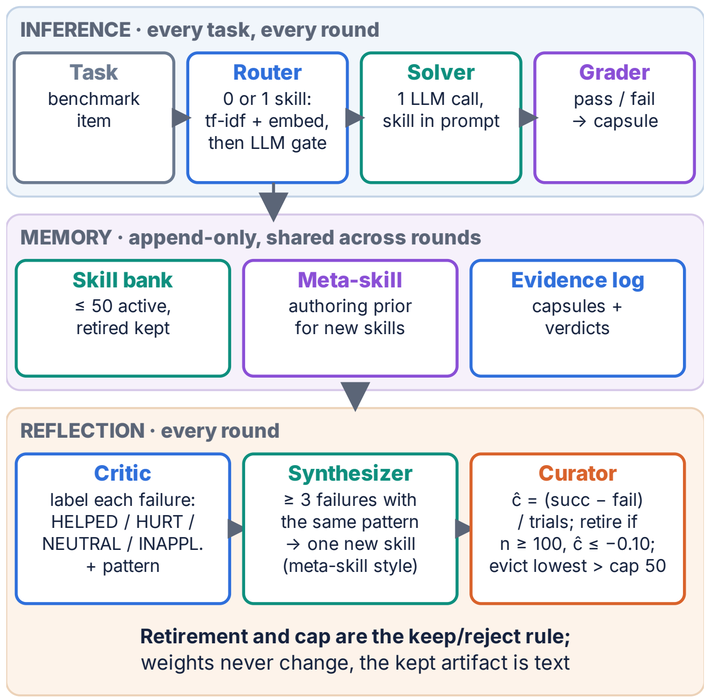
HarnessRSI 74 ★★★

evidence: retained

Paper

Code

1 What changes, what is kept



2 Gain by condition (MBPP+ hard-100)

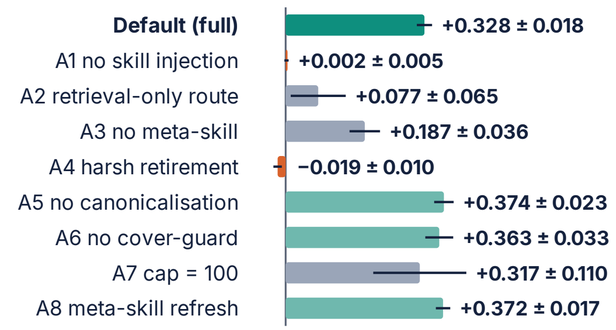


Table 1, mean ± std over 3 seeds; gain = last-10-round mean − first-10-round mean of held-out pass@1 (40 eval tasks). Default: baseline 0.258 → late-window mean 0.584, peak 0.658. Light green: above the Default.

3 Controls that decide if the gain is real

No-skill floor and ablations (3 seeds each)

- **A1** skills never injected: +0.002 (but Table 1 lists 0 critic calls and 42 active skills: inconsistent)
- **A3** no meta-skill: +0.187 (0.141 below Default)
- **A4** N_min 100 → 20 *and* τ 0.10 → 0.0 (two knobs, not separated): bank collapses to 2 skills

Gain per seed (Table 3)	42	7	13
Default	+0.303	+0.340	+0.343
A1 no injection	+0.003	**−0.005**	+0.008
A2 retrieval-only	+0.148	+0.095	**−0.010**
A4 harsh retire	**−0.005**	**−0.027**	**−0.025**
A7 cap = 100	+0.172	+0.340	+0.440

One fixed-seed 60/40 split (Sec. 6.1); round-0 baselines vary 0.175–0.350 across runs. Every A8 seed (+0.365, +0.355, +0.395) sits above every Default seed.

Cost and not run

- ~14.5k calls vs 10k (A1), same solver calls; 6.5 h vs 2.3 h
- **Not reported:** other systems as baselines, compute-matched no-skill run, leakage check; whether eval outcomes feed the Curator

4 Where the gain is absent or thin

- **No row removes retirement or the cap**; no unbounded-library run. The drift below the floor (A4, −0.019) comes from *harsher* retirement, and is less than one eval task (0.025‡)
- A5, A6, A8 (+0.374, +0.363, +0.372) score **above the Default** (+0.328); the paper: explicit dedup is not necessary
- A2 +0.077 ± 0.065 and A7 ± 0.110: small or noisy. Peak is noisy too: A1 never injects, yet peaks at 0.375 (baseline 0.283)
- One benchmark, one model, 40 eval tasks; no external skill-library baseline; Prop. 1 floor E[p0] − 0.35 is "loose"

Takeaway

One 40-task coding suite, one model: a governed skill library gains +0.328, the same loop with injection off +0.002, and the meta-skill is the most valuable of the ablated components. Whether unbounded growth causes drift, or retirement and the cap are essential, is not directly tested.

Our read · HarnessRSI

Score 74 · ★★★ · evidence level: retained

Reports: held-out split, 3 seeds, 8 ablations, cost.
Does not report: transfer, other-system baselines, matched compute, leakage check.